ISSUE #02

Monthly Newsletter

CATALYST
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JANUARY 2026

EDITOR'S NOTE

Here's what has happened in the last month and what's to come!

Welcome to the January edition of the Catalyst Newsletter. This publication captures ongoing technical progress, project milestones, and engineering efforts undertaken by Catalyst as a student-led think tank focused on building, executing, and delivering meaningful technical systems. Over the past month, our teams have continued working across diverse domains, translating ideas into functioning hardware, software, and autonomous systems.

Firstly, we would like to express our gratitude to our Principal, Dr. P.B. Mane and our faculty coordinator Prof. Venkat Ghodke. Your generosity and dedication have allowed us to keep our tasks running smoothly and effectively.

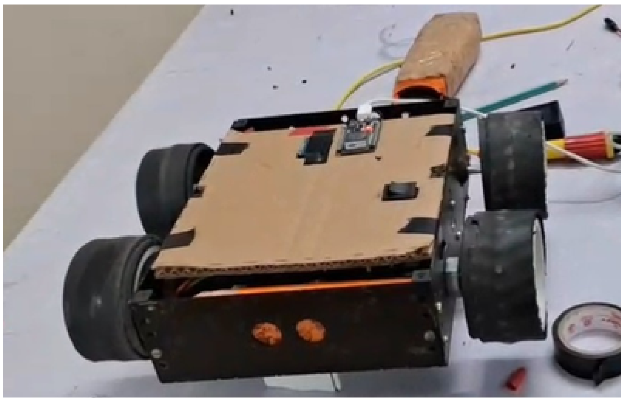
This edition highlights progress across multiple ongoing projects, including autonomous aerial systems, embedded hardware development, robotics and SLAM-based navigation, open-source flight control infrastructure, documentation and knowledge management platforms, and low-cost indigenous laboratory equipment. Alongside technical development, we also share insights from competitive platforms and engineering challenges that have shaped our approach to system design, testing, and execution.

Finally, if you are interested in contributing ideas, collaborating with our teams and contributing to projects that actually matter, we welcome you to reach out.

Enjoy this month's newsletter!
Created by Yashada Rane

Catalyst Team x

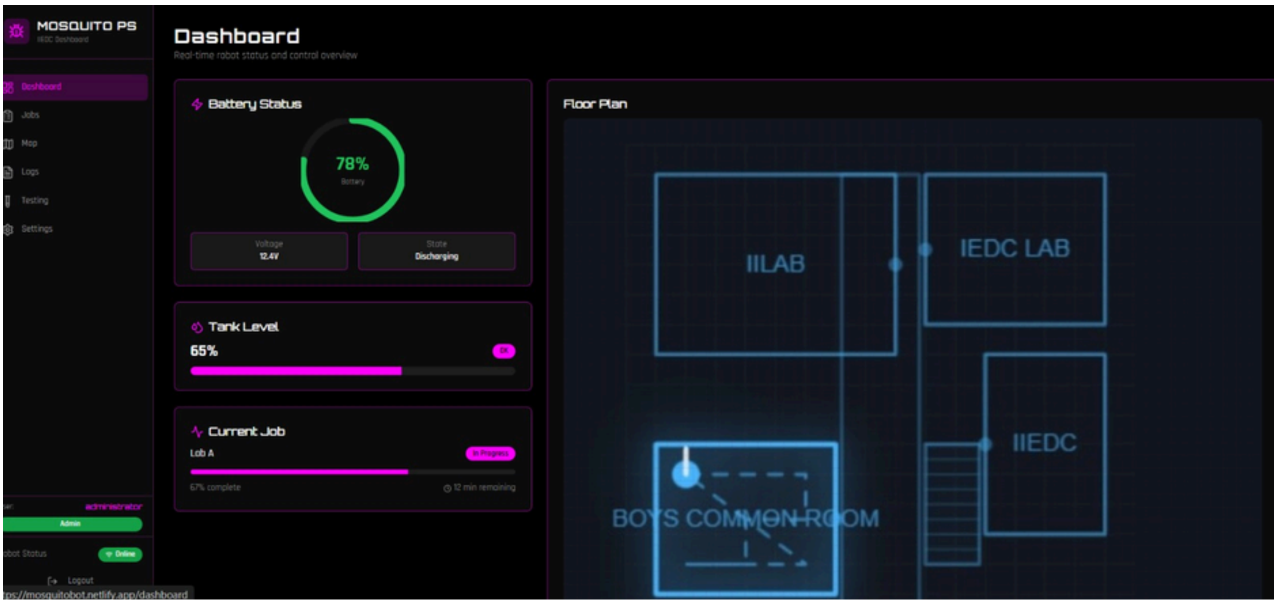
In this newsletter you can expect:

Building an Autonomous Indoor Robot Using SLAM and LiDAR

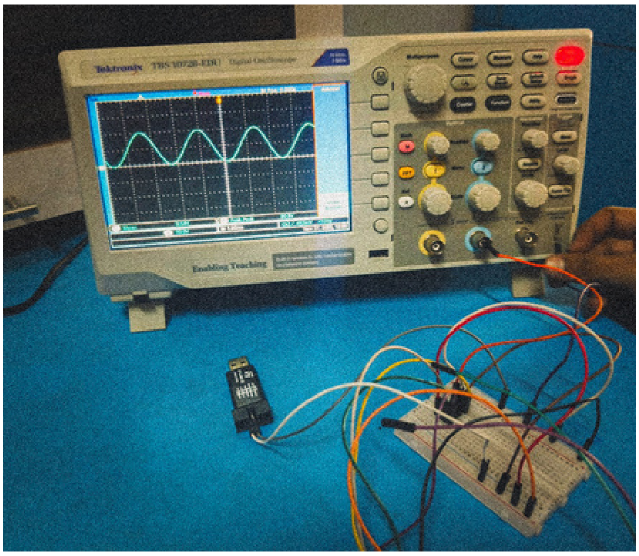
The project led by Aaron Mobby is focused on developing an indoor mobile robot capable of transitioning from web-based control to fully autonomous navigation using SLAM and onboard perception. The robot is currently operated through a web interface, with motion commands transmitted over an MQTT server to an ESP32 responsible for motor control. Autonomous navigation is being implemented on a Raspberry Pi 4B using ROS2 Jazzy as middleware, with real-time environmental perception provided by a rotating TF-Luna LiDAR sensor. The required SLAM and robotics software packages have been installed and configured, establishing a functional mapping and localization framework. LiDAR integration via serial communication has been completed, and accuracy testing has confirmed millimeter-level precision for distances ranging from 20 cm to 8 m, making it a reliable and cost-effective sensor for indoor mapping.

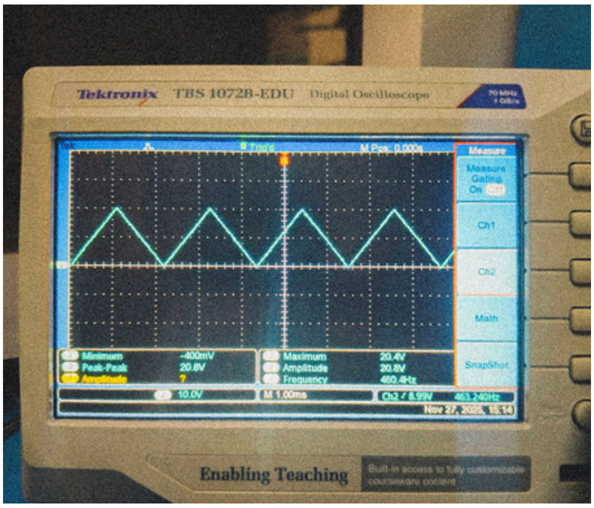
Earlier development stages successfully generated occupancy grid maps using Python-based visualization, and work is now underway to fully integrate map generation within the ROS2 SLAM framework. The LiDAR sensor performs multi-directional scans through servo-based motion, with sequential scan data streamed directly into the SLAM pipeline for real-time localization and loop closure detection. At present, robot movement is commanded via the web interface, while ongoing development aims to shift decision-making fully onboard the Raspberry Pi. This includes generating motion commands internally based on SLAM outputs and executing them through serial communication to the ESP32, along with implementing frontier-based exploration to enable autonomous navigation of previously unvisited regions.



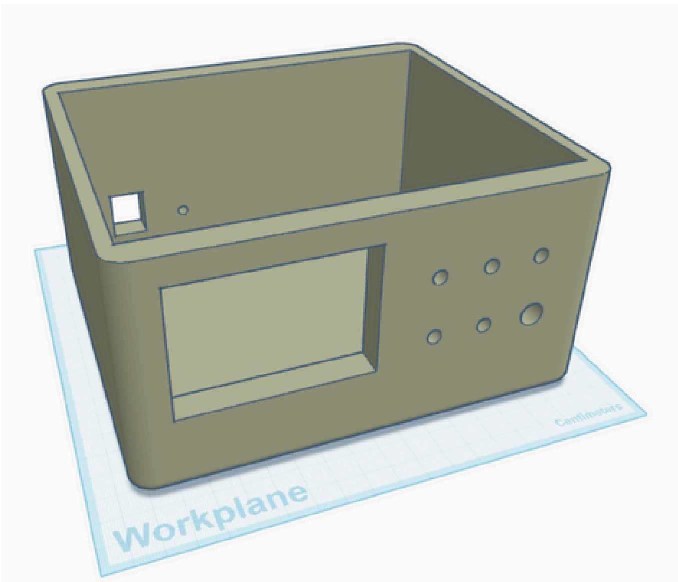
Final Milestones in STM32 Function Generator Development

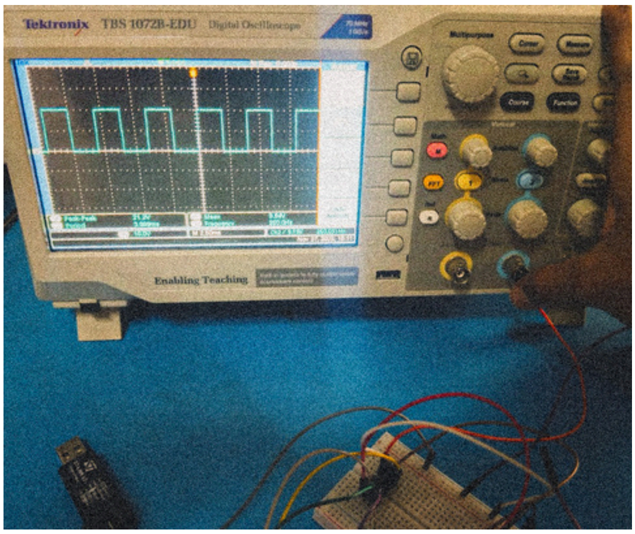
The STM32-based single-channel Function Generator project has progressed to its final stage, with the system now 95% complete. All core hardware and firmware components have been successfully designed, assembled, integrated and thoroughly tested. The device reliably generates sine, square and triangle waveforms across a 1 Hz to 100 kHz frequency range with adjustable 0–5 V peak-to-peak output, DC offset and duty cycle control. Performance validation using an oscilloscope has confirmed waveform accuracy, frequency stability, amplitude linearity and acceptable harmonic distortion across the operating range. The TFT display and rotary encoder based user interface are fully functional, enabling intuitive real-time control of waveform parameters with smooth and responsive feedback.



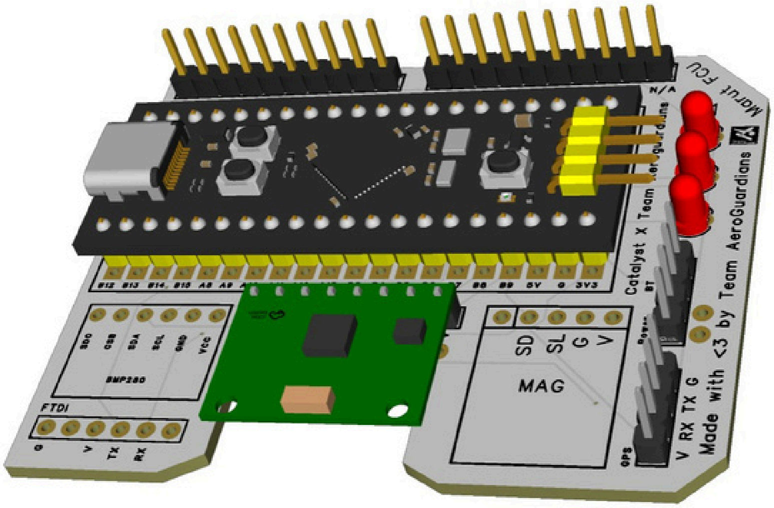


The system architecture combines digital waveform synthesis on the STM32F411 microcontroller with a precision 12-bit DAC and analog reconstruction filtering, ensuring clean and stable output suitable for laboratory use. At present, the only remaining step is final assembly and enclosure integration. The 3D enclosure design has been completed, and the team is currently awaiting competitive quotations to proceed with 3D printing and final housing. Once the enclosure is assembled, the project will be ready for demonstration and evaluation. Led by Arya Nirhali, with Sanket Kolhe and Vedant Nalawade, the project represents a robust indigenous alternative to costly commercial function generators, achieving significant cost reduction while maintaining reliable performance for educational and experimental applications.





Marut V1 Flight Controller PCB: Design and Architecture Overview



This month's highlight is the custom PCB developed for the V1 launch of Marut, designed to transform rapid prototyping into a clean, modular, and reliable hardware platform. The board is intended to replace fragile wiring-heavy setups with a structured design that supports repeatable testing, easier debugging, and long-term development. From the outset, the focus was on building a PCB that could serve both as a learning platform and as a dependable foundation for real flight experimentation.

The PCB is built around a proven MCU development board footprint, allowing developers to leverage a familiar and well-supported processing platform while integrating flight-critical peripherals directly onto a single board. Essential components such as the IMU, barometer, GPS interface, and multiple expansion headers are cleanly integrated, eliminating the need for excessive jumper wiring. Connector placement has been carefully planned to maintain logical signal flow and accessibility, while clear silkscreen labeling across the board improves traceability during debugging and reduces onboarding friction for new contributors.

From a layout and electrical design perspective, the board prioritizes signal integrity, power stability, and noise isolation. Sensitive analog and digital sensor circuits are physically separated from high-current power paths and motor-related interfaces, reducing electromagnetic interference and improving measurement reliability. Wide power traces, proper grounding strategies, and intentional routing decisions help ensure stable operation under varying load conditions. Despite these considerations, the PCB maintains a compact and stackable form factor, making it suitable for integration into layered system architectures and future expansions.

The design also emphasizes accessibility and hackability. All major interfaces are exposed, debug points are easy to reach, and expansion headers allow for rapid prototyping of additional sensors or modules without redesigning the core board. Rather than locking users into rigid configurations, the PCB is intentionally flexible, encouraging experimentation and iteration. This balance between structure and openness reflects Marut's broader engineering philosophy: hardware should be robust enough to trust in the air, yet adaptable enough to evolve alongside the software and mission requirements.

Overall, the Marut V1 PCB represents a significant step toward making custom flight control hardware less cluttered, less fragile, and more disciplined. It lays a strong hardware foundation for upcoming flight tests and autonomous features, while remaining an approachable platform for students to learn, modify, and build upon with confidence.

Beyond Takeoff: What NIDAR Really Taught Us

Being selected as finalists among 350+ teams and representing our institute at NIDAR, one of India's largest drone competitions, was an achievement in itself. However, the experience quickly revealed that NIDAR was never just about flying from point A to point B. It was a rigorous test of engineering judgment, system-level thinking, and the ability to operate under real-world constraints where theory meets uncertainty.

Over the course of four months, our team worked through the complete lifecycle of an autonomous aerial system. This included hardware selection, embedded software development, sensor integration, navigation logic, communication pipelines, power management, and mission planning. Unlike controlled lab experiments or simulations, this process involved real hardware operating in unpredictable environments. Every decision carried consequences, and many assumptions were challenged once the system left the workbench and entered the field.

NIDAR pushed us to think beyond individual subsystems and focus on holistic system behavior. Reliable autonomy required us to handle noisy sensors, inconsistent GPS data, intermittent communication links, power limitations, and strict timing constraints. Several issues surfaced only after prolonged field testing, and others appeared for the first time once the system was fully assembled and airborne.

Debugging in such conditions demanded rapid decision-making, adaptability, and a strong understanding of how each subsystem influenced the others.

The competition also reinforced the importance of structured testing, redundancy, and defensive design. Despite months of validation, certain failures emerged only on mission day itself. These moments highlighted a fundamental engineering reality: systems rarely fail where you expect them to. Designing for uncertainty, edge cases, and graceful failure became just as critical as optimizing for performance and efficiency.

While we may not have walked away with a trophy, the experience delivered something far more valuable. We built and operated a functioning autonomous aerial system under constraints that closely resemble real-world deployments. We learned how to collaborate under pressure, make informed trade-offs when time and resources were limited, and recover from setbacks without losing momentum or focus.

Most importantly, NIDAR reminded us that engineering success is not defined solely by awards or rankings. It is defined by the depth of understanding gained, the ability to solve complex problems under uncertainty, and the lessons carried forward into future work. The insights from this journey will continue to shape how we design, test, and think as engineers long after the competition has ended.

Thank you for reading!